

Computer Science Department
Software Engineering & Business Analysis

Bachelor's Thesis

MovieCrush

Movie Tracking Platform with Personalized
Recommendations, Detailed Ratings, and Social Features

Anastasiia Syvak

Supervisor

KSE, Denys Zavhorodnii, dzavhorodnii@kse.org.ua

Expert

KSE, Expert Name, expert@domain.ua

Submission date

16 June 2026

TODOS

Academic Integrity Statement

I, undersigned, hereby declare that this capstone project is the result of my own work.

- All ideas, data, figures and text from other authors have been clearly cited and listed in the bibliography.
- No part of this project has been submitted previously for academic credit in this or any other institution.
- All code, diagrams, and third-party materials are either my original work or are used with permission and properly referenced.
- I have not engaged in plagiarism or any form of academic dishonesty.
- Any assistance received (e.g. from peers, tutors, or online forums) is acknowledged in the acknowledgements section.

I understand that failure to comply with these declarations constitutes academic misconduct and may lead to disciplinary action.

Place, date Kyiv, 03.06.2026

Signature _____

Contents

Acknowledgements	1
Abstract	3
1 Introduction	4
1.1 Context and Motivation	4
1.2 Objectives	4
1.3 Research Approach	5
1.4 Structure of This Thesis	5
2 Research	6
2.1 Section 1	7
2.2 Section 2	7
2.3 Conclusion	7
3 Design	8
3.1 Section 1	9
3.2 Section 2	9
3.3 Conclusion	9
4 Implementation	10
4.1 Section 1	11
4.2 Section 2	11
4.3 Conclusion	11
5 Validation	12
5.1 Section 1	13
5.2 Section 2	13
5.3 Conclusion	13
6 Conclusion	14
6.1 Project Summary	14
6.2 Comparison with Initial Objectives	14
6.3 Encountered Difficulties	14
6.4 Future Perspectives	15
Glossary	16

Acknowledgements

First of all, I want to thank my parents. For giving me the opportunity to study at such a great university, for their support every single time, for always believing in me when I didn't believe in myself - they always knew I would make it. They have always been my biggest role models in life, and everything I have now is because of them. I also want to thank my mom for being someone I can tell absolutely everything to - every secret, every worry, every thought. And I want to thank my dad for teaching me tennis when I was a child - that's what led me to become the head of the KSE Table Tennis Club and find amazing friends at university, who made these student years the best ones.

I want to thank my grandparents - they were always ready to help me with anything, and were even willing to learn how to code just to make things easier for me. Thank you for the fact that when I had classes until 9 PM, I could get to Akademmistechko in 20 minutes instead of an hour and a half, and thank you for always cooking so much and so well every time I came over, as if it was a wedding and not just my last class of the day. I want to thank my brother for always showing up - early or late - even though he wasn't a student here at that moment, just to help or support me.

I want to thank Veronika Anepska for absolutely everything - for always listening to me, for every party, every conversation, every idea, for believing in each other and supporting each other. Together we built discipline - not just opening TikTok every day to keep our streak alive, but also learning Italian every day, doing yoga every day, setting goals and reaching them together. I'm really happy that this diploma was made with you. I'm sure we have an amazing future ahead, because we're doing everything to make it happen.

I want to thank Bohdana Seheda for sitting next to me in an extra English class in first year and Maryna Chernetska for the fact that we were brought together in first year by the same problem - neither of us understood anything. Those small things helped me find such great friends.

Thank you to Ivan Suprun - after talking to you, everything always feels lighter. Thank you to Marko Hlibovytskyi for always telling me I would be fine when I felt like giving up, and for helping me install Windows when I needed it.

I really want to thank the EBD27 and BE27, and my Verkhovyna company. Thank you to Daria Kozlova for the Integration parties, for Halloween vampires, and for inviting me in my fourth year to get through the hard winter moments and the power outages at Hotel Verkhovyna together. Thank you to Dmytro Shevchuk, Matvii Onyshchenko, Artem Litskevych, Dmytro Vasylchuk, Oleksandr Siryi - you are the best. Every holiday, every party, every memory with you is one of the best in my life. Thank you for every tennis game, every good lunch, every conversation, You are all absolute legends. Thank you to Oleh Skakun for always cheering me up and for driving me home.

Thank you to my friend from Italy, Gabriel, whom I met just before the full-scale invasion started - for helping me feel more confident, for every Eurovision night, for respecting my family and my country, and for the invitation to Italy. Thank you also to my friend from Spain, Markel - I'm sure we will meet someday too.

Thank you to Polina Mamchur, Sofiia Tyshchenko, Sofiia Krotova, Yaroslava Mala and Kateryna Shashkina for every moment we went through together, for finding something positive even during the worst deadline weeks, and for always helping.

I also want to thank Kateryna Pyvovarova for our conversation on the 5th floor of KSE - it was you who came up with the Soulmate feature idea, and I'm really glad that conversation happened.

I want to thank the teachers who left a real mark. Thank you to Taras Pavlov - he taught mathematics in the first year and managed to make difficult topics understandable, and more than that - he made me love them even more. Thank you to Yevhen Kubiuk for the practical sessions on algorithms and data structures: I did not miss a single class, because you rarely meet someone who loves what they do so genuinely and wants every student to truly understand. Thank you to Yaroslav Propoi for the practical product analytics course - it was after this course that I understood I want to build my life around analytics. Thank you to Vadym Yaremenko and Ihor Pysmennyi for their huge contribution to my knowledge. Thank you to Volodymyr Skochko - he returned my assignments with new comments every single time, and at some point I genuinely did not know when it would end. But at work I understood what he was preparing me for: clients are often unsatisfied, there are many rounds of revisions, and you just have to keep going. I am ready for that now. And a separate thank you for agreeing to help me install a virtual Windows on my Linux - even though it did not work out, what matters is that he did not say no and tried to help. Thank you to academic director Artem Korotenko - over four years our leadership changed several times, and Artem inherited a situation that was not easy to walk into. He handles it well, and more importantly - you can always come to him with a question and get real help or at least honest advice. That is not something to take for granted. And a very big thank you to my supervisor Denys Zavorodnii. For every meeting, for the time spent reviewing my work, for the guidance, and for walking me through this entire process from start to finish. It was a real pleasure working with you.

Thank you to absolutely everyone who came into my life during these years and made it the best it could be - in such difficult times.

Abstract

The abstract serves as a concise summary of your entire thesis, encapsulating key elements on a single page such as:

- *General background information*
- *Objective(s)*
- *Approach and method*
- *Conclusions*

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

1 | Introduction

1.1 Context and Motivation

I have been obsessed with films and series for as long as I can remember. My friends know that if they want to talk about cinema, I am always available. Films help me switch off, step into another world, experience someone else's problems, and discover something new. If I had grown up in California instead of Ukraine, I would probably have studied acting, not computer science.

This personal connection is also what made me notice the problems with existing tools. For two years I used both Letterboxd and TV Time, and over that time I found enough frustrations to fill a product backlog. I started with phone notes - a list of films I wanted to watch, and eventually moved to Letterboxd because I wanted to actually track what I had seen and follow friends to see what they were watching. But even there, things were missing. No series support. No personalized recommendations. Statistics locked behind a paywall. No Ukrainian language. And the social layer felt thin.

The decision to build MovieCrush came from a simple idea: instead of renting someone else's apartment and living with their design choices, build your own - with exactly the layout you always wanted. I am the target audience of this product, which made every design and product decision easier to reason about. I was not building for an imaginary user. I was building for myself and people like me.

The broader problem MovieCrush addresses is one that a survey of 262 respondents confirmed: 49.8% of people track films in phone notes, and 83.6% discover content through social media rather than dedicated platforms. Existing solutions each cover only part of the user journey - Letterboxd handles films but ignores series, TV Time covers series but has weak recommendations and no social transparency, and IMDb is a reference database rather than a personal tracking tool. The core question "what should I watch tonight?" remains poorly answered by all of them.

1.2 Objectives

The goal of this capstone project is to design, build, and evaluate a complete cross-platform mobile application that unifies film and TV tracking, personalized recommendations, social discovery, and personal analytics in a single product.

The specific objectives are:

- Build a fully functional mobile app for Android (with iOS compatibility) using React Native and Expo
- Implement a hybrid recommendation engine combining ALS collaborative filtering with TMDB Discover candidate generation and Google Gemini re-ranking
- Design and evaluate a Soulmate matching algorithm that finds users with the most similar taste using a weighted combination of five similarity metrics
- Build a Wrapped feature that generates personalized yearly viewing statistics

- Validate the recommendation model using standard information retrieval metrics and evaluate the product through real user testing

1.3 Research Approach

Before writing a single line of code, three complementary research methods were applied: a quantitative survey of 262 respondents, five qualitative in-depth interviews, and a structured competitor gap analysis covering Letterboxd, IMDb, TV Time, and Moviebase. This ensured that product decisions were grounded in real user needs rather than assumptions.

Development followed an iterative sprint model over three months. The recommendation system went through five optimization iterations - each starting with a problem, forming a hypothesis, running an experiment, and measuring the result. This approach is documented in full in the Validation section.

1.4 Structure of This Thesis

The thesis is organized as follows:

- **Analysis** - user research findings, competitor gap analysis, business model, and domain-specific technical challenges
- **Design** - system architecture, technology stack decisions, database schema, capacity planning, and load estimation
- **Implementation** - development sprints, UX design, key algorithms (Soulmate similarity and hybrid recommendation pipeline), and deployment setup
- **Validation** - unit testing, ML model evaluation across five iterations, manual testing with real user feedback, requirements validation, and limitations

2 | Research

In this section, you perform a State of the Art investigation of your domain to understand what's been built, how it works, and where it falls short. You'll survey existing solutions—tools, frameworks and approaches—then compare them against your project's requirements to justify why a new tool (and its specific feature set) is needed.

A focused Domain Area Research unfolds in these steps:

- 1. Clarify your research questions and functional requirements*
- 2. Collect and review candidate solutions*
- 3. Evaluate each alternative's strengths, weaknesses, maturity and architectural style*
- 4. Group similar approaches into meaningful categories (for example, monolithic vs. microservice, commercial vs. open-source)*
- 5. Pinpoint gaps or missing features that motivate your own design*

This process carries your initial concept through systematic research all the way to a clear, actionable set of requirements for your proposed tool.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

Contents

2.1 Section 1	7
2.2 Section 2	7
2.3 Conclusion	7

2.1 Section 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

2.2 Section 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

2.3 Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

3 | Design

In this section you turn your requirements into a concrete engineering blueprint. You'll justify every major architectural choice, visualize structure with C4 diagrams for the first three layers, and map out your runtime topology so that peers can understand—and you can defend—every aspect of your system.

1. Clarify how functional and non-functional requirements drive your high-level architecture
2. List each architectural decision (for example, “We chose microservices to enable independent scaling and deployment”) and explain why it best meets your goals
3. Include a C4 Context diagram showing your system in its environment (users, external systems, data sources)
4. Include a C4 Container diagram breaking the system into deployable units (APIs, web front end, background workers, databases) and annotate communication styles and protocols
5. Include a C4 Component diagram for your core container(s), illustrating key modules, services or libraries and their interactions
6. Describe your deployment topology: physical or cloud hosts, network zones, load-balancing, failover and backup strategies
7. Summarize your technology stack, mapping each tool or framework back to a specific container or component and noting any trade-offs (performance, community support, learning curve)
8. Outline how data flows through the system—including storage models, messaging patterns or API contracts—and note any schema or interface versioning plans
9. Address cross-cutting concerns (Security, Logging, Monitoring, Scalability) and show where they sit in your topology

By walking through Requirements → Decisions → Diagrams → Topology, your Design section becomes a rigorous, evidence-backed foundation for the implementation that follows.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Quod idem licet transferre in voluptatem, ut.

Contents

3.1 Section 1	9
3.2 Section 2	9
3.3 Conclusion	9

3.1 Section 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

3.2 Section 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

3.3 Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

4 | Implementation

In this section you translate your component-level designs into working code and systems. Focus on the C4 Component layer and on the details needed to show how your design was realized. Include only the most important code snippets that illustrate key patterns or algorithms, rather than full listings.

1. *Describe the development methodology (for example, Agile or test-driven development) used to guide your implementation*
2. *Explain any prototyping or iterative strategies you applied to refine components before full-scale coding*
3. *Summarize coding standards, naming conventions and architectural patterns followed in your codebase*
4. *Present critical code snippets or configuration templates that highlight how core components were implemented (for example, key classes, interfaces or algorithms)*
5. *Detail your testing approach and quality assurance measures (unit tests, integration tests, coverage metrics)*
6. *Note any performance optimizations or profiling results for components that were bottlenecks*
7. *Outline your deployment and configuration management process for component artifacts (containerization, CI/CD pipelines)*
8. *Highlight documentation deliverables (API references, inline comments, architecture decision records) that support future maintenance*

This section demonstrates how each component specification becomes actual, maintainable code—closing the loop from design to implementation.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Quod idem licet transferre in voluptatem, ut.

Contents

4.1 Section 1	11
4.2 Section 2	11
4.3 Conclusion	11

4.1 Section 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

4.2 Section 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

4.3 Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

5 | Validation

Validation (Requirements Verification and Testing)

In this section you demonstrate how your implementation satisfies the initial requirements through clear testing methods and concise examples—suitable for a bachelor-level project:

1. *Restate each key functional and non-functional requirement from your Analysis and Design sections*
2. *Describe the testing approach for each requirement (for example, unit tests, manual acceptance checks or scenario walkthroughs)*
3. *Provide concrete test cases or usage examples that show how you verify each requirement in practice*
4. *Summarize actual versus expected outcomes, indicating pass/fail status for each test*
5. *Include brief snippets of test code or sample console outputs to illustrate your procedures*
6. *Note any gaps or deviations and suggest simple fixes or areas for future improvement*
7. *If a feature wasn't intended for specific scenarios (e.g. high-load), omit unrealistic stress tests and clearly document its current limitations*

This focused structure ties every requirement directly to validation results, using examples and methods you can realistically carry out at the bachelor level.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Quod idem licet transferre in voluptatem, ut.

Contents

5.1 Section 1	13
5.2 Section 2	13
5.3 Conclusion	13

5.1 Section 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

5.2 Section 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

5.3 Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

6 | Conclusion

6.1 Project Summary

MovieCrush is a fully functional cross-platform mobile application that unifies film and TV tracking, personalized recommendations, social discovery, and personal analytics in a single product. The system was built over three months as a solo project, resulting in a React Native mobile client, a Node.js and Express backend with 11 domain modules, a PostgreSQL database with 17 tables, and a separate Python microservice running an ALS collaborative filtering model.

The recommendation engine is a three-stage hybrid pipeline: ALS collaborative filtering generates candidates, TMDB Discover adds variety, and Google Gemini 2.5 Flash re-ranks the combined pool and assigns each result a category - strong match, diversity, or hidden gem. For new users, a dedicated cold start service uses onboarding signals to serve initial recommendations until enough watch history accumulates for ALS. The Soulmate feature finds the user with the most similar taste using a weighted combination of five metrics: rating cosine similarity, watched overlap, actor overlap, mood similarity, and disliked overlap. The Wrapped feature generates personalized yearly statistics across top genres, directors, actors, moods, and total watch time.

The ALS model was evaluated across five optimization iterations on a synthetically generated dataset, reaching a final ROC AUC of 0.8771 - substantially above the random baseline of 0.5. Three external users tested the app and provided feedback that directly influenced product decisions and surfaced a performance bug that was found and fixed during the session.

6.2 Comparison with Initial Objectives

The core objectives set at the start of the project were met. A fully functional Android app was built and distributed as an APK. The hybrid recommendation pipeline combining ALS, TMDB Discover, and Gemini was implemented and evaluated. The Soulmate algorithm was built, tested, and deployed. Wrapped analytics were implemented and well received by testers - one tester compared it directly to Spotify Wrapped and said they would love to share their film stats in Instagram Stories.

Two planned features were not implemented within the project timeline: Challenges and gamification, and push notifications for premiere dates. Both were consciously deprioritized after reviewing survey results, which showed that Wrapped (rated 4.24 out of 5), AI recommendations, and social features were the top priorities for the target audience. Public iOS distribution was not completed due to the cost of an Apple Developer account, though the app is cross-platform compatible.

6.3 Encountered Difficulties

The hardest part of this project was the recommendation system - not technically, but in terms of decision-making. The approach changed four times over the course of development.

The first plan was to use an AI chat assistant similar to ChatGPT but limited to film topics. This was dropped early in favor of a single AI assistant button that generates personalized recommendations using DeepSeek. DeepSeek was then dropped after the committee pointed out that a purely LLM-based approach could ignore large prompts, lose context when given many films, or return irrelevant results. Nine models were tested with five different prompts each - Claude Opus 4.7, Claude Haiku 4.7, Claude Sonnet 4.6, GPT 5.4, GPT 5.4 mini, GPT 5.5, Gemini Flash, Gemini Pro, and Gemini Thinking - before the same concern was confirmed: a pure LLM approach was too unpredictable for personalization at scale.

The final solution was a hybrid pipeline where collaborative filtering does the heavy lifting and Gemini acts only as a re-ranker on a pre-filtered candidate pool. This architecture is more reliable, more explainable, and produces better results - but it took four iterations of rethinking to get there. Looking back, this iterative process of forming a hypothesis, testing it, and being willing to throw it away and start again was the most valuable engineering lesson of the whole project.

Other notable challenges included the AWS account being blocked due to a billing issue mid-project; the Neon + Render split database setup adding latency and exposing the database to the public internet, which was solved by moving everything to Render Postgres; and a `director_similarity` metric in the Soulmate algorithm that was computing against a table that nothing ever populated, silently burning 10% of the weight budget until it was caught in a code review and removed.

6.4 Future Perspectives

The most immediate next steps are automated ALS model retraining on a cron schedule, and a custom domain for Resend to enable full email verification for all users rather than just the account owner.

On the product side, the features that testers and survey respondents asked for most are: Instagram Stories share templates for Wrapped and Soulmate results, a permanent country filter in recommendations, streaming availability information on film pages (showing which platform a film is available on), and Challenges with achievements and viewing goals.

On the business side, the planned monetization path is a freemium model with a MovieCrush Pro subscription at 1.99 USD per month on the annual plan, unlocking AI recommendations, Soulmate search, and no ads. Affiliate partnerships with streaming platforms in the style of JustWatch are also planned as an additional revenue stream.

Longer term, as the real user base grows and the interaction matrix becomes larger and denser, the ALS model is expected to improve significantly - the current evaluation was conducted on a synthetic dataset of 208 users, and real behavioral patterns at scale will provide much richer training signal. At that point, online evaluation with A/B testing would replace the current offline metric approach and give a more accurate picture of recommendation quality in production.

Glossary

Bibliography

A | Appendix

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.